Automated Fish Species Classification

Using Sonar and Machine Learning

Shiheng Huang

Objectives

Develop automated workflow for ecological monitoring in Ontario lakes with hydroacoustic data

Proposed Models

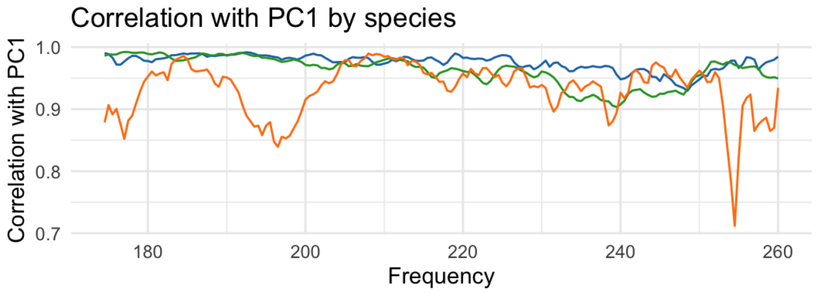
- Multinomial Regression:
  - ➡ First two PCs (Cum. Var.: 95.9%)
- Random Forest:
  - ➡ Mtry: 20, n_trees: 400, min_n = 5
- Deep Neural Network:
  - ➡ 3 hidden layers
- Long Short-Term Memory:
  - ➡ 2 LSTM + 1 hidden layers

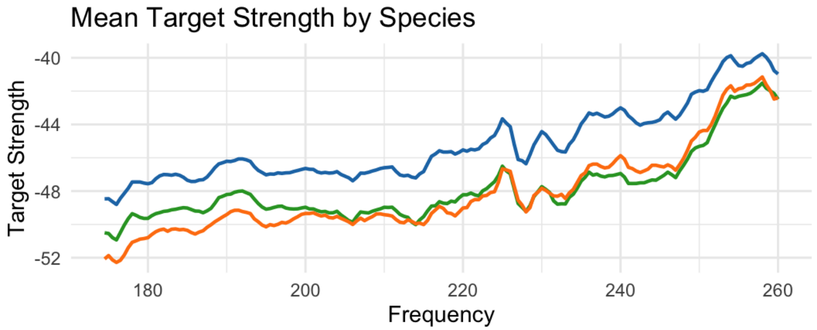
Future Steps

- LSTM with attention mechanism
- Incorporate Fish Characteristics

Datasets

- 14,392 observations, 264 predictors
- 3 fish species:
  - Lake Trout (21), Lake Whitefish(10),
  - Smallmouth Bass (12)
- Frequency Range:
  - F45 - F89.5, F173 to F260
- 80% Train (11802 obs), 20% Test (2590 obs)





Results

- Classified on fish-level
- LSTM best captures interactions
  - Multinomial Regression (37.5%)
  - Random Forest (62.5%)
  - DNN (62.5%)
  - ✦ LSTM (62.5%)

LSTM Confusion matrix

Actual / Predicted	LT	LWT	SMB
LT	2	0	1
LWT	1	2	0
SMB	1	0	1

- Only LSTM detects all three species
- Species habit pattern is crucial
- Need to find best models' structures